

Tightening Welfare Identification Under Nonconvex Preferences

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Abstract

Revealed-preference welfare bounds that rely on convexity of preferences are invalid when preferences may be nonconvex; the monotonicity-only bounds of Halevy et al. (2017) remain valid but are typically wide. We characterize the identifying content of two structural restrictions that do not require convexity. A finite dataset is rationalizable by a symmetric and homothetic utility function if and only if the symmetrized dataset, obtained by adjoining every permutation of each price–quantity pair, satisfies the homothetic axiom of revealed preference; symmetry and homotheticity therefore reduce to a single test on an expanded dataset rather than to two separate conditions. A homothetic analogue of the monotone revealed-preference relation then yields inner and outer bounds on every admissible strict upper contour set, valid uniformly over the class of symmetric homothetic rationalizations and hence over nonconvex members of that class. The induced expenditure intervals contract the monotonicity-only benchmark weakly, and strictly when an added comparison is expenditure-relevant at the counterfactual price; when the two endpoints coincide, reference-contour expenditure is point identified within the maintained class.

Keywords: Revealed preference; symmetry; homothetic utility; nonconvex preferences; welfare bounds.

JEL Classification: D11, D12, C14, C60.

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1 Introduction

The recovery of welfare-relevant objects from observed choice data lies at the core of revealed preference theory. Classical contributions, beginning with Samuelson (1938) and culminating in Varian (1982, 1983), provide sharp nonparametric characterizations of utility maximization and deliver informative bounds on counterfactual demand and welfare. A central ingredient behind these results is *convexity of preferences*, which guarantees the existence of supporting prices and permits revealed-preference inequalities to tightly discipline indifference curves and expenditure functions.

In many economically relevant environments, however, convexity is implausible. Indivisibilities, discrete choices, behavioral frictions, and kinked or piecewise-linear preferences naturally generate nonconvexities. Without convexity, the supporting-price argument breaks down and the classical convexity-based bounds can be invalid. Halevy et al. (2017) therefore construct alternative bounds based only on monotonicity; those bounds are valid for nonconvex preferences but are generally wider. This motivates the central question of the present paper: *can symmetry and homotheticity jointly restore welfare identifying power in nonconvex settings, and if so to what extent, measured by the reduction in welfare uncertainty relative to the monotone baseline?*

The present paper establishes the affirmative. The key observation is that symmetry and homotheticity each propagate revealed-preference comparisons across a distinct dimension of the choice space: symmetry through permutations of goods, homotheticity through rescaling along rays from the origin. Each propagation generates revealed-preference chains absent from the original dataset. Such chains are *expenditure-relevant* when they change the cost-active part of the inner or outer contour bound at the counterfactual price; in that case the corresponding welfare endpoint strictly tightens. This is not the generic observation that any additional restriction weakly shrinks the contour-uncertainty region. The substantive contribution of the paper is to show that symmetry and homotheticity generate expenditure-relevant chains that *strictly* tighten the bounds, and to characterize the limiting case in which the bounds collapse to a singleton.

The analysis proceeds in two stages. The first stage characterizes when symme-

try and homotheticity are consistent with the observed demand data, yielding an empirically testable necessary and sufficient condition. The second stage determines what those restrictions deliver for welfare identification conditional on the test being satisfied. The two stages are logically dependent: Lemma 3.8 is the prerequisite for the welfare analysis, since the bounds in Sections 4 and 5 are informative only when the joint hypothesis is not refuted by the data.

The relationship between convexity, symmetry, and homotheticity as identification devices warrants clarification. Three considerations favor symmetry and homotheticity over convexity in the environments studied here.

First, domain of applicability. Convexity requires upper contour sets to be convex, a condition violated under indivisibilities, kinked preferences, and discrete choice. Symmetry and homotheticity, by contrast, do not impose convexity. They impose permutation invariance and radial replication of preference rankings and are therefore compatible with nonconvex upper contour sets, as illustrated in Examples 3.3–3.5. These restrictions do not eliminate the source of nonconvexity; rather, they provide an alternative route to welfare identification precisely when convexity fails.

Second, empirical testability. Both symmetry and homotheticity admit sharp revealed-preference characterizations from finite choice data without auxiliary functional-form assumptions: homotheticity is characterized by HARP (Varian, 1983), and symmetry by the conditions of Christopher and Rehbeck (2018). GARP characterizes the existence of a well-behaved concave rationalization, but it does not establish that the consumer’s true preferences are convex. The issue emphasized by Halevy et al. (2017) is instead that welfare bounds constructed from supporting prices can fail when a nonconvex rationalization is the true one.

Third, economic motivation. Symmetry is a strong restriction on preferences over distinct commodities, and we do not defend it there. It is, however, the standard maintained assumption in a class of problems in which the coordinates of the choice vector are not commodities at all but *states, dates, assets, or persons*, none of which carries intrinsic weight independently of the quantity attached to it. Four such settings recur, and nonconvexity is common in all of them.

In *state-contingent consumption* with exchangeable states, a decision maker who regards the states as ex ante indistinguishable ranks (c_1, c_2) and (c_2, c_1) alike; sym-

metry is then implied by exchangeability rather than assumed on top of it, while indivisibilities and probability weighting routinely generate nonconvex upper contour sets. In *portfolio choice over homogeneous assets*, an investor holding 100 units of asset A and 50 of asset B is indifferent to the reverse holding when the assets are identical in risk and return (Fudenberg et al., 2015). In *time-dated consumption* of a non-storable good under time-additive utility with identical period felicity and no discounting, the consumer assigns equal utility to $(c_t, c_{t'})$ and $(c_{t'}, c_t)$ (Matějka and McKay, 2015). In *social evaluation over anonymous individuals*, anonymity is precisely permutation invariance, and the objects being ranked are allocations rather than bundles.

Two features of these environments are worth recording together. The coordinates are numerous, so L is typically large; and symmetry there is not a parametric convenience but a substantive restriction, refutable through the conditions of Christopher and Rehbeck (2018). The first feature makes the factorial size of the symmetrized dataset a genuine obstacle, which is why Proposition 3.11(iii) supplies a screen that resolves most pairs in $O(L \log L)$; the second is what makes the exercise informative when the screen does not reject. Homotheticity, defined as the scale invariance of preference rankings, is appropriate when total expenditure affects the scale but not the composition of consumption, a maintained assumption supported empirically in aggregate and household demand (Varian, 1983; Heufer and Hjertstrand, 2019).

While symmetry and homotheticity do not restore the supporting-price argument of convexity, they generate an alternative source of identifying power by propagating revealed-preference comparisons across permutations of goods and scales of consumption, systematically contracting welfare bounds without imposing convexity on indifference sets.

The paper establishes four results, taken up in turn.

Lemma 3.8 characterizes joint rationalizability: a finite dataset admits a rationalization by a symmetric and homothetic utility function if and only if its symmetrized expansion O_Σ satisfies HARP. HARP on the original dataset alone omits cycles that combine permuted observations. The joint hypothesis therefore requires HARP on O_Σ , which enforces all permutation-induced comparisons simultaneously and yields a single test for the two restrictions. Condition (v) of the lemma shows that the Afriat inequalities take a particularly compact form under the joint hypothesis, facilitating

computation. A pairwise restriction extracted from that system, PHARP, refutes the joint hypothesis whenever it fails, though passing it does not establish rationalizability, since longer cycles can still violate HARP (Proposition 3.11). It therefore serves as a one-sided screening device, and a permutation reduction together with a rearrangement bound resolves most pairs in $O(L \log L)$ rather than $O(L!)$, which matters because the settings where symmetry is credible are those with many coordinates.

Theorems 4.6 and 5.3 deliver the welfare bounds. When the joint hypothesis is not refuted, symmetry and homotheticity yield valid nonparametric inner and outer bounds on strict upper contour sets and on reference-contour expenditure. Symmetry alone tightens the monotonicity-based bounds of Halevy et al. (2017) (Proposition 4.2); homotheticity provides further contraction by propagating comparisons along rays. The bounds are strictly narrower whenever the additional comparisons are expenditure-relevant at the counterfactual price (Corollary 5.4). Theorem 5.3 establishes uniform validity over the maintained class; it does not assert that the endpoints are attainable by admissible rationalizations.

Theorem 5.5 isolates the limiting case. When the valid lower and upper expenditure endpoints coincide, reference-contour expenditure is exactly identified within the maintained class despite nonconvexity. A common-minimum condition stated in terms of the outer set and the closure of the inner set provides a verifiable sufficient condition.

Proposition 5.7 quantifies the gain. The bound-tightening index $\kappa(p, x) := 1 - W^{SH}(p, x)/W^M(p, x) \in [0, 1]$ measures the fractional contraction of the displayed structural expenditure interval relative to the monotonicity-only baseline of Halevy et al. (2017). It vanishes when the restrictions leave that interval unchanged and equals unity when the structural interval collapses. Because sharpness is not established, the index is interpreted as a diagnostic for this bounding procedure rather than as the fraction of the sharp identified set resolved by the maintained assumptions.

Relation to the literature. A growing literature has explored alternative sources of identifying power in revealed-preference analysis. The present paper contributes to several strands of this literature.

Nonconvex preferences. Halevy et al. (2017) establish that the monotonicity-only

contour bounds remain valid in nonconvex settings but can be wide. Their bounds constitute the baseline against which the results of the present paper are measured: symmetry and homotheticity weakly narrow those bounds, and do so strictly when the symmetrized expansion generates a new comparison that is cost-active at the counterfactual price. Nishimura et al. (2017) develop a comprehensive approach to revealed preference under minimal rationality assumptions; the present framework imposes more structure and in return delivers tighter valid welfare bounds.

Stochastic and population revealed preference. Kitamura and Stoye (2018) and Aguiar et al. (2023) study revealed preference in stochastic settings, where identification is obtained from distributional restrictions rather than from individual choice consistency. The approach of the present paper is complementary: attention is restricted to a single consumer’s observed choices, and identification is derived from structural restrictions on the utility function rather than from distributional assumptions.

Homotheticity. Varian (1983) establishes the revealed-preference characterization of homotheticity that underlies the present analysis. Heufer and Hjertstrand (2019) develops computational methods for testing and measuring departures from homotheticity. Echenique (2014) studies the computational difficulty of testing separability, while Cherchye et al. (2015) develops a revealed-preference and integer-programming approach to weak separability. The contribution of the present paper relative to this strand is twofold: symmetry and homotheticity are combined to derive joint revealed-preference conditions (Lemma 3.8), and this combination is shown to restore identifying power in nonconvex environments not considered in the cited literature.

Engel curve literature. A complementary strand tightens welfare bounds by combining revealed-preference restrictions with shape restrictions on Engel curves. Blundell et al. (2003) show that Engel curve restrictions substantially narrow the contour-uncertainty region for welfare changes; Blundell et al. (2007) and Blundell et al. (2008) extend these results to semiparametric settings and derive sharp consumer-surplus bounds; Blundell et al. (2015) derive sharp SARP-consistent bounds on counterfactual demand and welfare; and Cherchye et al. (2020) show how normal-goods restrictions tighten cost-of-living analysis in panel data. These papers exploit income variation across observations, whereas the approach of the present paper is applicable to a single cross-section and relies on functional-form restrictions rather than shape

restrictions on Engel curves. The two approaches are complementary and admit combination when both income variation and structural restrictions are available.

Symmetric rationalizability. Christopher and Rehbeck (2018) establish necessary and sufficient conditions for rationalizability by a symmetric utility function, and Chambers and Echenique (2016) provide a comprehensive axiomatic treatment of structural restrictions in revealed preference theory. Neither paper derives welfare bounds in nonconvex settings, nor combines symmetry with homotheticity to exploit the joint identification power characterized in Lemma 3.8. The contribution of the present paper is to show how symmetry functions as a welfare-identification device and how its interaction with homotheticity generates revealed-preference content that is invisible when the two restrictions are analyzed separately.

Symmetry arises naturally in models of rational inattention and information processing (Fudenberg et al., 2015; Matějka and McKay, 2015), in portfolio allocation with interchangeable assets, and in aggregate demand settings where symmetry is a maintained modeling assumption, providing additional motivation for the environments studied here.

Computational feasibility. The cardinality of the symmetrized dataset O_Σ grows as $T \cdot L!$, which may be large when L is not small. Once the expanded dataset is formed, pairwise PHARP requires only $T^2 L!$ comparisons, since the (σ, η) grid collapses onto the single composition $\pi = \eta \circ \sigma^{-1}$ (Proposition 3.11(ii)); a rearrangement bound resolves most ordered pairs in $O(L \log L)$ without any enumeration (part (iii) of the same proposition). PHARP provides a necessary-condition screen: any failed pair immediately refutes the joint hypothesis.¹ Passing PHARP is not sufficient, because a longer HARP cycle may still fail; the full cyclic test must then be solved. The procedure is polynomial in the size of the explicitly symmetrized dataset but factorial in the primitive number of goods. Repeated coordinates and permutation symmetries may reduce the number of distinct comparisons in particular applications.

The two stages of the analysis are logically dependent. Section 3 is the empirical prerequisite for Sections 4 and 5: the welfare bounds are informative only when the joint hypothesis satisfies the HARP condition on O_Σ . When that condition fails, the

¹The pairwise conditions are formalized as the Pairwise Homothetic Axiom of Revealed Preference (PHARP) in Proposition 3.11.

maintained assumptions are refuted and the bounds are vacuous. The structure of the paper reflects this dependency: the empirical license is established before it is exploited.

The remainder of the paper is organized as follows. Section 2 introduces the environment and the principal definitions. Section 3 presents the revealed-preference characterization of the joint hypothesis. Sections 4 and 5 derive bounds on indifference curves and the expenditure function under the maintained hypothesis. Section 6 concludes. All proofs are collected in the Appendix.

2 Preliminaries

Suppose that a consumer chooses bundles consisting of $L \geq 2$ goods. We observe a finite number T of price-quantity pairs, indexed by $\mathbb{T} = \{1, 2, \dots, T\}$. Let $y_t = (y_t^1, \dots, y_t^L) \in X \equiv \mathbb{R}_+^L$ denote the bundle chosen at observation $t \in \mathbb{T}$, purchased at prices $p_t = (p_t^1, \dots, p_t^L) \in P \equiv \mathbb{R}_{++}^L$.² We impose Walras' law throughout: wealth at observation t equals $p_t y_t$, and assume $p_t \cdot y_t > 0$ for every observation. We denote by $O = \{(p_t, y_t)\}_{t \in \mathbb{T}}$ the dataset of all price-quantity observations.

Let Σ denote the set of all permutations $\sigma : \{1, \dots, L\} \rightarrow \{1, \dots, L\}$. Given (p_t, y_t) , define the permuted price and bundle vectors

$$p_t^\sigma = (p_t^{\sigma(1)}, \dots, p_t^{\sigma(L)}) \quad \text{and} \quad y_t^\sigma = (y_t^{\sigma(1)}, \dots, y_t^{\sigma(L)}),$$

and the permuted dataset $O_\sigma = \{(p_t^\sigma, y_t^\sigma)\}_{t \in \mathbb{T}}$. Concatenating across all $\sigma \in \Sigma$ yields the *symmetrized dataset*

$$O_\Sigma = \bigsqcup_{\sigma \in \Sigma} O_\sigma = \{(p_t^\sigma, y_t^\sigma)\}_{(t, \sigma) \in \mathbb{T} \times \Sigma}.$$

By construction, O_Σ contains $T \times |\Sigma| = T \times L!$ observations and any test or statistic based on O_Σ is invariant to relabeling of goods.

²We use the following notation: $\mathbb{R}_+^L = \{y \in \mathbb{R}^L : y \geq (0, \dots, 0)'\}$ and $\mathbb{R}_{++}^L = \{y \in \mathbb{R}^L : y > (0, \dots, 0)'\}$. Note that $p_t y_t = \sum_{l=1}^L p_t^l y_t^l$.

Example 2.1 (Symmetrized dataset). Let $L = 2$, $\mathbb{T} = \{1, 2\}$, and

$$O = \{((1, 2), (2, 1)), ((3, 1), (1, 2))\}.$$

With $\Sigma = \{\text{id}, \tau\}$ where τ swaps the two coordinates, the symmetrized dataset is

$$O_\Sigma = \{((1, 2), (2, 1)), ((3, 1), (1, 2)), ((2, 1), (1, 2)), ((1, 3), (2, 1))\}.$$

Each observation together with its coordinate-swapped counterpart appears in O_Σ .

To formalize these ideas, we begin with the standard revealed-preference relations that will serve as a benchmark for the structural extensions developed below.

Definition 2.2 (Direct and Indirect Revealed Preference). *The bundle y_t is directly revealed preferred to y_s , written $y_t \succeq^{R,D} y_s$, if $p_t y_t \geq p_t y_s$; and strictly directly revealed preferred, written $y_t \succ^{R,D} y_s$, if $p_t y_t > p_t y_s$.*

The bundle y_t is revealed preferred to y_s , written $y_t \succeq^R y_s$, if there exists a chain $y_t \succeq^{R,D} y_{t_1} \succeq^{R,D} \dots \succeq^{R,D} y_s$.

Definition 2.3 (GARP). *The dataset O satisfies the Generalized Axiom of Revealed Preference (GARP) if there is no cycle $y_t \succeq^R y_s$ and $y_s \succ^{R,D} y_t$.*

GARP characterizes rationalizability by a well-behaved utility function. The homothetic and symmetry-based conditions introduced below impose additional structure through scale invariance and permutation symmetry without requiring convexity of the consumer's true upper contour sets.

2.1 Revealed Preference under Homotheticity

The key idea is that preference comparisons are invariant to radial rescaling of bundles under homothetic preferences.

Definition 2.4 (Homothetic Revealed Preference). *Let $\mathcal{O}$ be any finite dataset of price-quantity pairs and let $x, z \in X$.*

- (i) *An observed bundle $(p_t, y_t) \in \mathcal{O}$ directly reveals y_t homothetically preferred to z , written $y_t \succeq_H^{R,D} z$ (resp. $y_t \succ_H^{R,D} z$), if $p_t \cdot y_t \geq p_t \cdot z$ (resp. $>$). This is*

the standard direct revealed-preference condition: z was affordable when y_t was chosen.

- (ii) An unobserved bundle $x \in X$ is directly revealed homothetically preferred to z , written $x \succeq_H^{R,D} z$, if there exist $(p_t, y_t) \in \mathcal{O}$ and $\lambda > 0$ such that $x = \lambda y_t$ and

$$p_t \cdot y_t \geq p_t \cdot \left(\frac{z}{\lambda}\right),$$

equivalently $p_t \cdot x \geq p_t \cdot z$. The relation is strict when the displayed affordability inequality is strict. The scale correction is essential: optimality first compares y_t with z/λ , and homotheticity then compares $\lambda y_t = x$ with z .

The bundle x is revealed homothetically preferred to z , written $x \succeq_H^R z$, if there is a finite chain $x \succeq_H^{R,D} y_{t_1} \succeq_H^{R,D} \dots \succeq_H^{R,D} z$ through observed bundles in $\mathcal{O}$; strictly ($x \succ_H^R z$) if at least one link is strict.

In the welfare analysis below, $\mathcal{O}$ is taken to be the symmetrized dataset O_Σ ; Definition 2.6 then follows by combining these homothetic relations with monotone dominance.

2.2 Monotonic Revealed Preference

The following recalls Halevy et al. (2017)'s definitions on monotonically revealed preference relations.

Definition 2.5 (Monotonic Preference). *The bundle y_t is monotonically preferred to y_s , written $y_t \succeq^M y_s$, if $y_t^l \geq y_s^l$ for all l ; and strictly so ($y_t \succ^M y_s$) if $y_t^l > y_s^l$ for all l .*

Definition 2.6 (Monotonic Homothetic Revealed Preference). *For arbitrary $x, z \in X$:*

- (i) x is directly monotonically homothetically revealed preferred to z , written $x \succeq_H^{M,R,D} z$, if $x \succeq^M z$ or $x \succeq_H^{R,D} z$; strictly ($x \succ_H^{M,R,D} z$) if the corresponding monotone or homothetic direct relation is strict.
- (ii) x is monotonically revealed homothetically preferred to z , written $x \succeq_H^{M,R} z$, if the relation holds directly, or there exist observed bundles $(y_{t_1}, \dots, y_{t_m}) \in O_\Sigma$

such that

$$x \succeq_H^{M,R,D} y_{t_1} \succeq_H^{R,D} \cdots \succeq_H^{R,D} y_{t_m} \succeq_H^{R,D} z.$$

Strictly ($x \succ_H^{M,R} z$) if at least one relation in the chain is strict.

The next definitions extend standard monotonic revealed preference by incorporating homothetic scaling. While Halevy et al. (2017) impose monotonicity alone, homotheticity introduces additional structure that propagates revealed-preference comparisons along rays, thereby strengthening identification and yielding tighter welfare bounds.

Definition 2.7 (Monotonic Homothetic Revealed Preference Sets). *For a bundle $x \in X$, define:*

- (i) *the monotonically and homothetically revealed preferred set: $MRP_{O_\Sigma}^H(x) = \{y \in X : y \succ_H^{M,R} x\}$;*
- (ii) *the monotonically and homothetically revealed worse set: $MRW_{O_\Sigma}^H(x) = \{y \in X : x \succ_H^{M,R} y\}$;*
- (iii) *the not monotonically and homothetically revealed worse set: $NMRW_{O_\Sigma}^H(x) = X \setminus MRW_{O_\Sigma}^H(x)$.*

When the dataset is clear from context, we write $MRP^H(x)$, $MRW^H(x)$, and $NMRW^H(x)$. The analogous sets without the homotheticity superscript, $MRP_O(x)$ and $MRW_O(x)$, denote the monotone revealed-preference sets of Halevy et al. (2017) constructed from the original dataset O .

3 Test of the Joint Hypothesis of Symmetry and Homotheticity

We begin by defining the two structural properties of preferences that are central to the paper.

Definition 3.1 (Homothetic Utility Function). *A utility function $u : X \rightarrow \mathbb{R}$ is homothetic if, for all $x, y \in X$ and all $\alpha > 0$,*

$$u(x) \geq u(y) \implies u(\alpha x) \geq u(\alpha y).$$

Definition 3.2 (Symmetric Utility Function). *A utility function $u : X \rightarrow \mathbb{R}$ is symmetric if, for any $x \in X$ and any permutation $\sigma \in \Sigma$,*

$$u(x^1, \dots, x^L) = u(x^{\sigma(1)}, \dots, x^{\sigma(L)}).$$

Homotheticity implies that preference rankings are invariant to proportional expansions of consumption bundles, so optimal choices scale proportionally with income. Symmetry requires the consumer to be indifferent to permutations of goods, treating all goods as intrinsically identical.

We illustrate with three examples of utility functions that are symmetric, homothetic, and nonconvex.

Example 3.3 (Symmetric L_r utility). *For $r > 1$, let*

$$u_r(x) = \left(\sum_{i=1}^K x_i^r \right)^{1/r}.$$

This utility is symmetric, strictly increasing, and homogeneous of degree one. Its upper contour sets are nonconvex: with two goods, $u_r(1, 0) = u_r(0, 1) = 1$, while $u_r(1/2, 1/2) = 2^{1/r-1} < 1$.

Example 3.4 (Max-plus CES). *Let*

$$u(y) = \max \left\{ \left(\sum_{i=1}^K y_i^p \right)^{1/p}, c \min_i y_i \right\}, \quad p > 1,$$

with $K^{1/p} < c < K$. Both components are symmetric and homogeneous of degree one, so the utility is homothetic. Both branches are active. Moreover, $u(e_i) = 1$ for each unit vector, whereas at their average $\bar{e} = (1/K, \dots, 1/K)$,

$$u(\bar{e}) = \max\{K^{1/p-1}, c/K\} < 1,$$

so the upper contour set at level one is nonconvex.

Example 3.5 (Strictly monotone perturbation). *For $r > 1$ and $\varepsilon > 0$, let*

$$u(x) = \left(\sum_{i=1}^K x_i^r \right)^{1/r} + \varepsilon \sum_{i=1}^K x_i.$$

The utility is symmetric, strictly increasing, and homogeneous of degree one. It remains non-quasiconcave: with two goods, $u(1, 0) = u(0, 1) = 1 + \varepsilon$, while $u(1/2, 1/2) = 2^{1/r-1} + \varepsilon < 1 + \varepsilon$.

Definition 3.6 (Homothetic Axiom of Revealed Preference). *The dataset O_Σ satisfies HARP if, for every sequence of indices $(i, j, k, \dots, l)$ in $\mathbb{T} \times \Sigma$,*

$$(p_i y_j)(p_j y_k) \cdots (p_l y_i) \geq (p_i y_i)(p_j y_j) \cdots (p_l y_l).$$

Note that Definition 3.6 states the standard HARP condition (Varian, 1983). When applied below to the *symmetrized* dataset O_Σ , it becomes a strictly stronger condition than HARP on the original dataset O , because O_Σ contains all permuted relabelings of the observed bundles. It is this application of HARP to O_Σ , not HARP on O alone, that jointly characterizes homotheticity and symmetry. The distinction matters: a dataset can satisfy HARP on O while failing HARP on O_Σ , as the numerical example in Section 3 illustrates.

We first recall Varian's nonparametric characterization of homothetic utility maximization, which provides the foundation for our results.

Theorem 3.7 (Varian, 1983). *For a dataset $O = \{(p_t, y_t)\}_{t \in \mathbb{T}}$, the following are equivalent:*

- (i) *There exists a nonsatiated homothetic utility function rationalizing O .*
- (ii) *O satisfies HARP.*
- (iii) *There exist numbers $U_t > 0$ such that*

$$U_s \leq U_t \frac{p_t \cdot y_s}{p_t \cdot y_t} \quad \text{for all } s, t \in \mathbb{T}.$$

- (iv) *There exists a continuous, strictly increasing, concave, degree-one homogeneous utility function rationalizing O .*

The next lemma is the first original result of the paper. It provides a full revealed-preference characterization of rationalizability by a utility function that is both homothetic and symmetric.

Lemma 3.8 (Joint Symmetry–Homotheticity Characterization). *Let*

$$O = \{(p_t, y_t)\}_{t \in \mathbb{T}}.$$

Let O_Σ denote its symmetrized expansion. The following are equivalent:

- (i) *There exists a continuous, strictly increasing, concave, degree-one homogeneous, and symmetric utility function rationalizing O .*
- (ii) *There exists a locally nonsatiated, homothetic, and symmetric utility function rationalizing O .*
- (iii) *The symmetrized dataset O_Σ satisfies HARP.*
- (iv) *There exist numbers $U_t^\sigma > 0$ such that, for all $s, t \in \mathbb{T}$ and $\sigma, \eta \in \Sigma$,*

$$U_s^\eta \leq U_t^\sigma \frac{p_t^\sigma \cdot y_s^\eta}{p_t^\sigma \cdot y_t^\sigma}.$$

- (v) *There exist numbers $U_t > 0$ such that, for all $s, t \in \mathbb{T}$ and all $\sigma, \eta \in \Sigma$,*

$$U_s \leq U_t \frac{p_t^\sigma \cdot y_s^\eta}{p_t^\sigma \cdot y_t^\sigma}.$$

Proof. See Appendix A. □

Remark 3.9 (On the concavity in part (i)). *Part (i) asserts that some concave rationalization exists whenever the test is passed. It does not assert that the consumer’s preferences are convex, and it does not license the classical convexity-based welfare bounds. This is the distinction emphasized by Halevy et al. (2017): a finite dataset consistent with utility maximization always admits a concave rationalization, yet bounds constructed from supporting prices of that particular rationalization can*

fail for a nonconvex member of the same admissible class. The admissible class $\mathcal{U}$ of Definition 5.1 accordingly contains nonconvex utilities, and every bound in Sections 4 and 5 is required to hold uniformly over that class, which is the content of Theorem 5.3. Part (i) is therefore a statement about what the data cannot rule out, not about what the consumer is like.

Lemma 3.8 does more than apply the homothetic and symmetry conditions separately to the original observations. Symmetry acts as a *dataset-expansion operator*: any utility that is both symmetric and homothetic must rationalize every permuted observation, so the relevant object is the symmetrized dataset O_Σ . HARP must be imposed on that expanded object, including cycles that combine different permutations. The nontrivial content of the lemma is that this condition is also *sufficient*, yielding a utility that is simultaneously symmetric, homothetic, continuous, and strictly increasing. Condition (v) provides a further payoff: the Afriat inequalities collapse to a single utility index $\{U_t\}$ satisfying scaled-price inequalities across all permutations simultaneously. This compact structure enables the welfare analysis of Sections 4 and 5: because all permuted comparisons are encoded in one system, revealed-preference propagation through O_Σ is computationally equivalent to testing HARP on an expanded dataset rather than solving a non-standard joint system.

Because HARP on the symmetrized dataset O_Σ is necessary and sufficient for joint symmetric-homothetic rationalizability (Lemma 3.8), any violation of HARP immediately refutes the joint hypothesis. In particular, exhibiting a violation at the level of a *length-2 cycle* is by itself sufficient for refutation, although length-2 cycles do not exhaust the violations HARP can detect. This observation yields a simple pairwise test that can be applied directly to the symmetrized dataset.

Definition 3.10 (PHARP on the symmetrized dataset). *The symmetrized dataset O_Σ satisfies PHARP if and only if, for all $s, t \in \mathbb{T}$ and all permutations $\sigma, \eta \in \Sigma$,*

$$(p_t^\sigma y_s^\eta)(p_s^\eta y_t^\sigma) \geq (p_t^\sigma y_t^\sigma)(p_s^\eta y_s^\eta).$$

PHARP isolates the pairwise (length-2) restrictions implied by HARP on the symmetrized dataset. It requires that no pair of (possibly permuted) observations generates a revealed-preference reversal once both directions are evaluated. Because symmetry expands the dataset through all permutations, PHARP aggregates these

pairwise comparisons across all relabelings of goods. As a result, it captures the simplest violations of joint symmetry–homothetic rationality while remaining directly testable.

The next result formalizes the computational content of PHARP and shows that these pairwise conditions are sufficient for refutation and can be verified without solving the full HARP feasibility problem.

Proposition 3.11 (PHARP as a computational refutation test). *Consider a dataset $O = \{(p_t, y_t)\}_{t \in \mathbb{T}}$ and its symmetrized expansion O_Σ .*

- (i) *Sufficiency for refutation: If PHARP on O_Σ is violated, then O cannot be rationalized by any continuous, strictly increasing, concave, homothetic, and symmetric utility function.*
- (ii) *Permutation reduction: The $(L!)^2$ pairs (σ, η) generate only $L!$ distinct inequalities. Writing $\pi := \eta \circ \sigma^{-1}$,*

$$p_t^\sigma \cdot y_s^\eta = p_t \cdot y_s^\pi, \quad p_s^\eta \cdot y_t^\sigma = p_s \cdot y_t^{\pi^{-1}}, \quad p_t^\sigma \cdot y_t^\sigma = p_t \cdot y_t,$$

so PHARP for the pair (s, t) is equivalent to

$$\min_{\pi \in \Sigma} (p_t \cdot y_s^\pi) (p_s \cdot y_t^{\pi^{-1}}) \geq (p_t \cdot y_t) (p_s \cdot y_s). \quad (1)$$

PHARP can therefore be verified by checking $T^2 L!$ inequalities rather than $T^2 (L!)^2$, without solving a global feasibility problem.

- (iii) *Rearrangement screen: Let*

$$m(p, y) := \min_{\pi \in \Sigma} p \cdot y^\pi,$$

which by the rearrangement inequality is obtained by pairing the components of p in ascending order with those of y in descending order, at cost $O(L \log L)$. Then:

- (a) *if $m(p_t, y_s) m(p_s, y_t) \geq (p_t \cdot y_t) (p_s \cdot y_s)$, the pair (s, t) satisfies PHARP;*
- (b) *if $(p_t \cdot y_s^\pi) (p_s \cdot y_t^{\pi^{-1}}) < (p_t \cdot y_t) (p_s \cdot y_s)$ for any single π , the pair violates PHARP.*

Enumeration over Σ is required only for pairs resolved by neither test.

- (iv) *Computational implication: PHARP provides a polynomial-time refutation procedure in the size of the explicitly symmetrized dataset. Its worst-case cost remains factorial in the primitive number of goods through $|O_\Sigma| = TL!$, but by part (iii) each pair is resolved in $O(L \log L)$ except in the band between the two screens, and passing PHARP does not replace the full HARP check.*

A clarification on the logical status of PHARP is important. PHARP is a *sufficient condition for refutation*: if PHARP is violated, then HARP fails on O_Σ and the joint symmetry-homotheticity hypothesis is rejected. PHARP is *not a necessary and sufficient test* for the joint hypothesis. A dataset can satisfy all pairwise PHARP inequalities while still violating HARP on O_Σ through a longer cycle (length ≥ 3). Equivalently, passing PHARP does not guarantee that O is rationalizable by a symmetric homothetic utility function; only the full HARP check on O_Σ provides that guarantee.

The value of PHARP lies in its role as a *fast screening device*. In practice, PHARP can reject the model at low computational cost whenever a pairwise violation is present, before the full HARP check is run. This mirrors the role of pairwise GARP checks in the literature, which detect violations efficiently without being logically equivalent to the full GARP condition. For a formal statement of the sufficiency, necessity, and computational properties, see Proposition 3.11 and its proof in the Appendix.

This result complements the refutation test of Christopher and Rehbeck (2018) for symmetric rationalizability. While their approach exploits pairwise comparisons induced by single permutations on O , PHARP extends this logic to the joint hypothesis by incorporating all permutation-induced observations in O_Σ .

Example 3.12 (Numerical refutation via PHARP on O_Σ , $L = 2$, $T = 2$). *Consider the dataset*

$$O = \{((4, 1), (1, 2)), ((1, 3), (3, 1))\}.$$

This dataset satisfies GARP and is therefore rationalizable by a monotone utility function.

The symmetrized dataset O_Σ consists of all permutations of each observation. For

$L = 2$, $\Sigma = \{\text{id}, \tau\}$, where τ swaps the two coordinates. Thus, O_Σ contains:

$$\begin{aligned}(p_1^{\text{id}}, y_1^{\text{id}}) &= ((4, 1), (1, 2)), \\(p_1^\tau, y_1^\tau) &= ((1, 4), (2, 1)), \\(p_2^{\text{id}}, y_2^{\text{id}}) &= ((1, 3), (3, 1)), \\(p_2^\tau, y_2^\tau) &= ((3, 1), (1, 3)).\end{aligned}$$

To test PHARP, consider the pair $(t, s) = (1, 2)$ with $\sigma = \text{id}$ and $\eta = \tau$. The PHARP inequality requires

$$(p_1^\sigma y_2^\eta)(p_2^\eta y_1^\sigma) \geq (p_1^\sigma y_1^\sigma)(p_2^\eta y_2^\eta).$$

Compute each term:

$$(p_1^\sigma y_2^\eta) = (4, 1) \cdot (1, 3) = 7, \quad (p_2^\eta y_1^\sigma) = (3, 1) \cdot (1, 2) = 5,$$

$$(p_1^\sigma y_1^\sigma) = (4, 1) \cdot (1, 2) = 6, \quad (p_2^\eta y_2^\eta) = (3, 1) \cdot (1, 3) = 6.$$

Hence,

$$(p_1^\sigma y_2^\eta)(p_2^\eta y_1^\sigma) = 7 \times 5 = 35 < 36 = 6 \times 6 = (p_1^\sigma y_1^\sigma)(p_2^\eta y_2^\eta).$$

PHARP is violated. Therefore, HARP fails on O_Σ , and by Lemma 3.8, the dataset O cannot be rationalized by any continuous, strictly increasing, concave, homothetic, and symmetric utility function.

This example illustrates how a violation arises from a single pairwise comparison across permuted observations, even though the original dataset satisfies GARP. In particular, the violation is driven entirely by symmetry-induced comparisons, which are invisible in the original dataset.

4 Bounding the Indifference Curve

Varian (1982) proposes a nonparametric recovery method that partially identifies

preferences by bounding indifference curves between a revealed-preferred inner set and a not-revealed-worse outer set. Under convexity, these sets are tight. Under nonconvexity, Halevy et al. (2017) show that Varian’s bounds can be violated and propose replacing them with monotonicity-based bounds: the inner bound is the set of bundles componentwise above the reference, and the outer bound is the set of bundles not componentwise below it. These monotone bounds are always valid, but wide when the data are sparse or when bundles are not componentwise comparable. Figure 1 illustrates the situation for $L = 2$: in the white region between the inner and outer bounds, the indifference curve is unconstrained by monotonicity alone.

Clarification on the role of nonconvexity. A potential source of confusion deserves explicit discussion. Lemma 3.8 (the characterization of the joint hypothesis) delivers a *concave* rationalizing utility as one of its equivalent conditions. This may suggest that the welfare bounds derived below are identical to those obtained under convexity. They are not, for the following reason.

Lemma 3.8 says: if O_Σ satisfies HARP, then a concave symmetric homothetic rationalization *exists*. But the welfare bounds in this section are valid for *all* monotone, homothetic, symmetric utilities that rationalize O_Σ , whether concave or not. The bounds are valid over the entire class of such utilities, which includes the nonconvex examples in Examples 3.3–3.5. The fact that a concave member of this class exists is used only to guarantee that the HARP condition is not vacuous; it does not restrict the class over which the bounds are computed.

Put differently: Lemma 3.8 is an *existence* result (a concave rationalization exists if and only if HARP holds), while the welfare bounds in Theorem 4.6 and Proposition 5.2 are *identification* results (what can be said about welfare when the true preference is any member of the class, possibly nonconvex). The bounds are tighter than those of Halevy et al. (2017) precisely because symmetry and homotheticity restrict the class of admissible utilities, even when no convexity is assumed, by generating new revealed-preference comparisons through permutations and scales.

This section shows how symmetry and homotheticity systematically tighten those monotonicity-based bounds by generating new revealed-preference comparisons. The tightening is visible in Figure 1: the darker shaded region (homothetic inner bound) extends into territory where monotonicity provides no information, and the outer

bound curve cuts through regions where the monotone outer bound is flat. Throughout, O_Σ is assumed to satisfy HARP and to be rationalized by a continuous, strictly increasing, homothetic, and symmetric utility function u (which need not be concave).

Definition 4.1 (Strict Upper Contour Set). *For a homothetic and symmetric utility function u , the strict upper contour set of $x \in X$ is $P_u(x) = \{y \in X : u(y) > u(x)\}$.*

4.1 Symmetry alone tightens identification

We first establish that symmetry, without homotheticity, already tightens the welfare bounds of Halevy et al. (2017) by exploiting the additional revealed-preference implications of permuted observations.

Proposition 4.2 (Symmetry tightens monotone bounds). *Let $O = \{(p_t, y_t)\}_{t \in \mathbb{T}}$ satisfy GARP and let O_Σ be its symmetrized expansion. For any monotone utility function u rationalizing O_Σ and any bundle x ,*

$$MRP_O(x) \subseteq MRP_{O_\Sigma}(x) \subseteq P_u(x) \subseteq NMRW_{O_\Sigma}(x) \subseteq NMRW_O(x),$$

and the first inclusion is strict at a reference bundle x whenever symmetrization generates a bundle y with $y \succ_{O_\Sigma}^{M,R} x$ but not $y \succ_O^{M,R} x$; the analogous condition on a newly revealed-worse bundle makes the final inclusion strict.

Proof. See Appendix A. □

Proposition 4.2 establishes that symmetry alone, without homotheticity, already tightens welfare bounds by exploiting the additional revealed-preference implications of permuted observations. The next results show that homotheticity provides further tightening.

4.2 Homotheticity expands the revealed-preference relation

Homotheticity adds valid radial comparisons to the monotonicity-only relation. On the original dataset O , the monotone relation of Halevy et al. (2017) is the special case $\lambda = 1$ of the corrected homothetic relation. Symmetrization then provides a second expansion by adding all permuted observations.

Proposition 4.3 (Homotheticity expands the revealed-preference relation). *For any bundle $x \in X$,*

$$MRP_O(x) \subseteq MRP_O^H(x) \quad \text{and} \quad MRW_O(x) \subseteq MRW_O^H(x),$$

equivalently $NMRW_O^H(x) \subseteq NMRW_O(x)$. Whenever a corrected radial comparison generates a chain absent from the monotonicity-only relation, the relevant inclusion is strict.

Proof. See Appendix A. □

Proposition 4.3 shows that every monotonicity-only chain remains available and that corrected radial comparisons may add new chains. The revealed-preferred and revealed-worse sets therefore expand, while the not-revealed-worse outer set contracts. Symmetrization compounds this effect by pooling all permutation-induced observations.

4.3 Bounds on the upper contour set under homotheticity

The following key lemma shows that homothetically revealed preference correctly orders utility.

Lemma 4.4 (Homothetic revealed preference implies utility ordering). *Let O_Σ satisfy HARP and let u be a continuous, strictly increasing, homothetic, and symmetric utility function rationalizing O_Σ . Then, for any $x, z \in X$,*

$$x \succ_H^{M,R} z \implies u(x) > u(z).$$

Proof. See Appendix A. □

Lemma 4.4 shows that the homothetic, monotone, and symmetric structure of preferences ensures that any bundle revealed preferred through these restrictions is strictly ranked higher in utility. In other words, the additional structural assumptions do not merely add comparisons; they also preserve the ordering implied by revealed preference. This provides a rigorous foundation for using symmetrized datasets to

derive tighter, nonparametric bounds on indifference curves and expenditure functions, as any bundle in the homothetically and symmetrically revealed-preferred set is guaranteed to yield higher utility under the rationalizing function.

Lemma 4.5 (Bounds on indifference curves under homotheticity). *Let O_Σ satisfy HARP and let u be a continuous, strictly increasing, homothetic, and symmetric utility function rationalizing O_Σ . Then, for every bundle $x \in X$,*

$$MRP_{O_\Sigma}^H(x) \subseteq P_u(x) \subseteq NMRW_{O_\Sigma}^H(x).$$

Proof. See Appendix A. □

Lemma 4.5 provides nonparametric upper and lower bounds on the true upper contour set when preferences are homothetic and the symmetrized dataset O_Σ is used. The inner bound $MRP_{O_\Sigma}^H(x)$ consists of bundles that are monotonically and homothetically revealed preferred to x through O_Σ ; these must yield strictly higher utility under any monotonic homothetic rationalization. The outer bound $NMRW_{O_\Sigma}^H(x)$ contains all bundles not monotonically and homothetically revealed worse than x ; no rationalizing utility can assign them utility strictly below $u(x)$.

4.4 Joint Action of Symmetry and Homotheticity

The following theorem is the main welfare-bounding result of the paper. It shows that the joint action of symmetry and homotheticity, operating through the symmetrized dataset O_Σ , yields a nested hierarchy of valid contour bounds. Strict tightening requires a new comparison on the relevant side of the contour bound.

Theorem 4.6 (Symmetry tightens homothetic revealed-preference bounds). *Let $O = \{(p_t, y_t)\}_{t \in \mathbb{T}}$ and let O_Σ be its symmetrized expansion. Suppose O_Σ satisfies HARP and is rationalized by a continuous, strictly increasing, homothetic, and symmetric utility function u . Then for every bundle x ,*

$$MRP_O^H(x) \subseteq MRP_{O_\Sigma}^H(x) \subseteq P_u(x) \subseteq NMRW_{O_\Sigma}^H(x) \subseteq NMRW_O^H(x).$$

Proof. See Appendix A. □

The sets $MRP_O^H(x)$ and $NMRW_O^H(x)$ are the homothetic monotonically-revealed-preferred and not-revealed-worse sets computed from the original dataset O ; $MRP_{O_\Sigma}^H(x)$ and $NMRW_{O_\Sigma}^H(x)$ are the analogous sets from O_Σ . Figure 1 illustrates the hierarchy geometrically for $L = 2$: the darker shaded region shows the inner bound $MRP_{O_\Sigma}^H(x)$, which strictly contains the monotone inner bound, while the outer bound $NMRW_{O_\Sigma}^H(x)$ is strictly smaller than its monotone counterpart. Because O_Σ is closed under a common permutation of prices and bundles, the resulting bounds are permutation-equivariant:

$$y \in MRP_{O_\Sigma}^H(x) \iff y^\sigma \in MRP_{O_\Sigma}^H(x^\sigma),$$

and analogously for MRW^H and $NMRW^H$. Symmetry therefore creates new chains by pooling all permuted observations; it does not require a comparison to hold separately under every permutation.

Corollary 4.7 (Strict tightening). *Under the conditions of Theorem 4.6, at least one inclusion is strict whenever O_Σ generates at least one homothetic revealed-preference relation not present in O . Specifically, if there exist permutations of observed bundles that produce a new chain $y \succ_H^{M,R} x$ in O_Σ but not in O , then*

$$MRP_O^H(x) \subsetneq MRP_{O_\Sigma}^H(x).$$

Analogously, a newly revealed relation of the form $x \succ_H^{M,R} y$ makes $NMRW_{O_\Sigma}^H(x) \subsetneq NMRW_O^H(x)$.

Proof. See Appendix A. □

Proposition 4.8 (Symmetry and homotheticity tighten monotone revealed preference). *For every $x \in X$,*

$$MRP_O(x) \subseteq MRP_{O_\Sigma}^H(x) \quad \text{and} \quad MRW_O(x) \subseteq MRW_{O_\Sigma}^H(x),$$

so that

$$NMRW_{O_\Sigma}^H(x) \subseteq NMRW_O(x).$$

Consequently, defining the contour-uncertainty bands

$$ID_{O_\Sigma}^H(x) := NMRW_{O_\Sigma}^H(x) \setminus MRP_{O_\Sigma}^H(x), \quad ID_O(x) := NMRW_O(x) \setminus MRP_O(x),$$

we have

$$ID_{O_\Sigma}^H(x) \subseteq ID_O(x).$$

Proof. See Appendix A. □

Proposition 4.8 formalizes the tightening effect of imposing symmetry and homotheticity on revealed preference analysis. Relative to the baseline monotone framework, the symmetrized dataset O_Σ generates additional revealed-preference relations, while homotheticity propagates these relations along rays. As a result, both the revealed-preferred and revealed-worse sets expand, the not-revealed-worse outer set contracts, and the contour-uncertainty band $ID_{O_\Sigma}^H(x)$ weakly shrinks relative to $ID_O(x)$.

The next example illustrates the identifying power of symmetry and homotheticity.

Example 4.9 (Identifying power of homotheticity). *Consider two goods and one observed choice*

$$p = (1, 2), \quad y = (3, 0),$$

from the budget $p \cdot z \leq 3$. The observation is rationalized, for example, by

$$u_\varepsilon(z) = \max\{z_1, z_2\} + \varepsilon(z_1 + z_2), \quad 0 < \varepsilon < 1,$$

which is symmetric, strictly increasing, homogeneous of degree one, and nonconcave. Let

$$x = 2y = (6, 0), \quad z = (5, 0.4).$$

The two bundles are not componentwise comparable, so monotonicity alone does not rank them. However,

$$p \cdot y = 3 > 2.9 = p \cdot (z/2),$$

which is the corrected radial affordability condition. Hence $x \succ_H^{R,D} z$ and every monotone homothetic rationalization ranks x strictly above z . The comparison is therefore added by homotheticity and is unavailable from the Halmi monotonicity-only relation.

5 Bounding the Expenditure Function

In this section, we show how the HALEVY-style contour bounds under homotheticity and symmetry deliver informative expenditure bounds. Let $u : X \rightarrow \mathbb{R}$ be a continuous, strictly increasing, homothetic, and symmetric utility function rationalizing O_Σ ; concavity is not imposed. For a counterfactual price vector $p \in \mathbb{R}_{++}^L$ and a reference bundle x , define the reference-contour expenditure

$$e_u(p; x) := \inf\{p \cdot z : u(z) \geq u(x)\}.$$

This welfare object is invariant to every strictly increasing transformation of u , so it refers to the same contour through x across all admissible ordinal representations.

5.1 Revealed-preference bounds

Let x be the reference bundle. Homotheticity does not imply that all bundles indifferent to x lie on one ray. Rather, under strict monotonicity each ray intersects a given indifference surface at most once, and different indifference surfaces are radial expansions of one another. The contour-uncertainty band

$$\mathcal{ID}_{O_\Sigma}^H(x) = NMRW_{O_\Sigma}^H(x) \setminus MRP_{O_\Sigma}^H(x)$$

contains every bundle that is not placed strictly above or strictly below x by the structural revealed-preference relation. We obtain two natural revealed-preference bounds:

$$\underline{e}(p, x) := \inf_{z \in NMRW_{O_\Sigma}^H(x)} p \cdot z, \tag{2}$$

$$\bar{e}(p, x) := \inf_{z \in MRP_{O_\Sigma}^H(x)} p \cdot z. \tag{3}$$

To compare the successive restrictions, write

$$\begin{aligned}
e_O^M(p, x) &:= \inf_{z \in NMRW_O(x)} p \cdot z, & \bar{e}_O^M(p, x) &:= \inf_{z \in MRP_O(x)} p \cdot z, \\
e_O^H(p, x) &:= \inf_{z \in NMRW_O^H(x)} p \cdot z, & \bar{e}_O^H(p, x) &:= \inf_{z \in MRP_O^H(x)} p \cdot z, \\
e_\Sigma^H(p, x) &:= \inf_{z \in NMRW_{O_\Sigma}^H(x)} p \cdot z, & \bar{e}_\Sigma^H(p, x) &:= \inf_{z \in MRP_{O_\Sigma}^H(x)} p \cdot z.
\end{aligned}$$

The unadorned endpoints $\underline{e}$ and $\bar{e}$ in (2)–(3) are $\underline{e}_\Sigma^H$ and $\bar{e}_\Sigma^H$. For the index below, set $\underline{e}^M := \underline{e}_O^M$, $\bar{e}^M := \bar{e}_O^M$, $\underline{e}^{SH} := \underline{e}_\Sigma^H$, and $\bar{e}^{SH} := \bar{e}_\Sigma^H$. Both sets are directly computable from revealed-preference relations using the symmetrized dataset.

Definition 5.1 (Uniform welfare bounds). *Let $\mathcal{U}$ denote the class of all continuous, strictly increasing, homothetic, and symmetric utility functions that rationalize O_Σ . For any $p \in \mathbb{R}_{++}^L$ and reference bundle x :*

- (i) $\underline{b}(p, x)$ is a uniformly valid lower bound if $\underline{b}(p, x) \leq e_u(p; x)$ for every $u \in \mathcal{U}$.
- (ii) $\bar{b}(p, x)$ is a uniformly valid upper bound if $e_u(p; x) \leq \bar{b}(p, x)$ for every $u \in \mathcal{U}$.

Uniform validity does not by itself imply sharpness. Sharpness would additionally require the endpoints to be attainable, or arbitrarily approachable, by admissible rationalizations.

Proposition 5.2 (Revealed-preference bounds on the expenditure function). *Let u be continuous, strictly increasing, symmetric, and homothetic, and assume u rationalizes O_Σ . For any $p \in \mathbb{R}_{++}^L$ and reference bundle x ,*

$$\underline{e}(p, x) \leq e_u(p; x) \leq \bar{e}(p, x).$$

Proof. See Appendix A. □

Theorem 5.3 (Uniform validity over the admissible class). *Let $\mathcal{U}$ be the class of Definition 5.1. Then*

$$\underline{e}(p, x) \leq \inf_{u \in \mathcal{U}} e_u(p; x) \leq \sup_{u \in \mathcal{U}} e_u(p; x) \leq \bar{e}(p, x).$$

Thus the two revealed-preference endpoints are uniformly valid over the entire admissible class. No equality with the infimum or supremum is asserted without an additional symmetric-homothetic order-extension result.

Proof. See Appendix A. □

The lower bound $\underline{e}(p, x)$ minimizes expenditure over the outer set $NMRW_{O_\Sigma}^H(x)$, which contains the strict upper contour set; minimizing over a larger set yields a smaller infimum. The upper bound minimizes over the inner set $MRP_{O_\Sigma}^H(x)$, which is contained in every admissible strict upper contour set; minimizing over a smaller set yields a larger infimum. Continuity and strict monotonicity imply that the infimum over the strict upper contour set equals the reference-contour expenditure $e_u(p; x)$.

5.2 Symmetry tightens expenditure bounds

The results of Section 4 establish that symmetry and homotheticity tighten the bounds on the upper contour set of x . These tighter set inclusions translate immediately into tighter bounds on the expenditure function. Specifically, enlarging the inner bound $MRP_{O_\Sigma}^H(x)$ weakly lowers the upper expenditure endpoint, while contracting the outer bound $NMRW_{O_\Sigma}^H(x)$ weakly raises the lower endpoint. The corollary below records the resulting chain of inequalities and characterizes when the tightening is strict.

Corollary 5.4 (Symmetry tightens expenditure bounds). *Figure 1 illustrates the tightening hierarchy: the red shaded region $MRW_{O_\Sigma}^H(x)$ identifies bundles homothetically revealed worse than x , with the solid curve separating that set from its complement $NMRW_{O_\Sigma}^H(x)$. The darker blue region $MRP_{O_\Sigma}^H(x)$ identifies bundles homothetically revealed preferred to x ; the true indifference curve through x is constrained to lie in the contour-uncertainty region between the two bounds.*

Under the assumptions of Theorem 4.6, symmetry enlarges the set of revealed-preferred bundles and shrinks the set of not-revealed-worse bundles:

$$MRP_O^H(x) \subseteq MRP_{O_\Sigma}^H(x), \quad NMRW_O^H(x) \subseteq NMRW_{O_\Sigma}^H(x).$$

Consequently, denoting bounds computed from O by a subscript O and from O_Σ by a

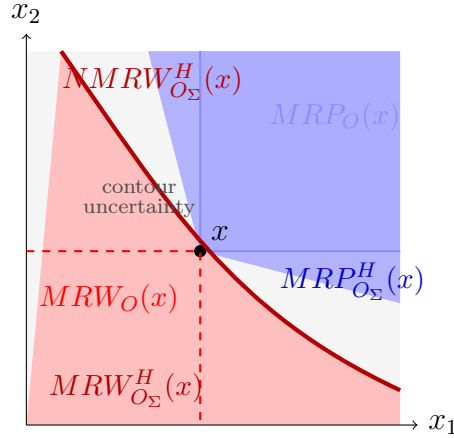


Figure 1: Tightening hierarchy for $L = 2$. The red shaded region $MRW_{O_\Sigma}^H(x)$ contains bundles homothetically revealed worse than x ; the solid red curve separates that set from its complement $NMRW_{O_\Sigma}^H(x)$ and lies strictly inside the monotonicity-only outer boundary (dashed L-shape). The darker blue region is the inner bound $MRP_{O_\Sigma}^H(x)$, which strictly contains the monotone inner bound $MRP_O(x)$ (lighter blue, northeast orthant). The contour-uncertainty region (between the solid red curve and the blue inner bound) contains all bundles admissible as indifferent to x ; the true indifference curve through x must lie entirely within it.

subscript Σ :

$$\underline{e}_O^M(p, x) \leq \underline{e}_O^H(p, x) \leq \underline{e}_\Sigma^H(p, x) \leq e_u(p; x) \leq \bar{e}_\Sigma^H(p, x) \leq \bar{e}_O^H(p, x) \leq \bar{e}_O^M(p, x).$$

Each successive expansion of the revealed relation weakly tightens the expenditure interval. A strict set expansion tightens an endpoint only when the new comparison changes the corresponding expenditure infimum.

Proof. See Appendix A. □

These bounds are tighter than those obtainable under homotheticity alone because the expanded inner set and contracted outer set exploit the joint structure of symmetry and homotheticity, ruling out a larger collection of observationally incompatible contour locations.

Theorem 5.5 (Identification collapse). *Let O_Σ satisfy HARP and be rationalized by a continuous, strictly increasing, homothetic, and symmetric utility function. Fix a reference bundle $x \in X$ and a price vector $p \in \mathbb{R}_{++}^L$.*

(i) If the frontier-contact condition holds,

$$\underline{e}_\Sigma(p, x) := \inf_{z \in NMRW_{O_\Sigma}^H(x)} p \cdot z = \inf_{z \in MRP_{O_\Sigma}^H(x)} p \cdot z =: \bar{e}_\Sigma(p, x),$$

then $e_u(p; x)$ equals this common value for every admissible rationalization u .

(ii) A sufficient condition is that the outer infimum is attained at some bundle z^* satisfying

$$z^* \in NMRW_{O_\Sigma}^H(x) \cap \overline{MRP_{O_\Sigma}^H(x)}.$$

Then both endpoints equal $p \cdot z^*$.

(iii) More generally, the frontier-contact condition holds if there exists e^* such that $p \cdot z \geq e^*$ for every $z \in NMRW_{O_\Sigma}^H(x)$ and, for every $\varepsilon > 0$, there is $w_\varepsilon \in MRP_{O_\Sigma}^H(x)$ with $p \cdot w_\varepsilon < e^* + \varepsilon$.

Proof. See Appendix A. □

The theorem shows that nonconvexity does not, by itself, preclude point identification of welfare. What matters is the richness of the revealed-preference information generated by symmetry and homotheticity. When these sources of propagation are sufficiently strong to align the inner and outer expenditure frontiers, the valid expenditure interval collapses to a singleton.

5.3 Quantifying the Magnitude of Bound Tightening

Results 1 and 2 establish that symmetry and homotheticity tighten welfare bounds whenever the symmetrized dataset generates expenditure-relevant chains. They do not, however, quantify the *magnitude* of this tightening. The following definition introduces a scalar index that answers this question for any dataset, price vector, and reference bundle, providing an observable and computable measure of the contraction of the displayed valid interval.

Definition 5.6 (Bound-tightening index). *Suppose all four expenditure endpoints are finite. Let $W^M(p, x) := \bar{e}^M(p, x) - \underline{e}^M(p, x) > 0$ denote the width of the expenditure*

interval under monotonicity alone, and let $W^{SH}(p, x) := \bar{e}^{SH}(p, x) - \underline{e}^{SH}(p, x)$ denote the width under symmetry and homotheticity. The bound-tightening index is

$$\kappa(p, x) := 1 - \frac{W^{SH}(p, x)}{W^M(p, x)} \in [0, 1].$$

The index measures the fractional contraction of the displayed Halevy-style expenditure interval relative to the monotonicity-only baseline. A value of $\kappa(p, x) = 0$ means the two procedures produce intervals of the same width. A value of $\kappa(p, x) = 1$ means the structural interval collapses and reference-contour expenditure is point identified within the maintained class. Without a separate sharpness theorem, intermediate values measure contraction of these particular valid bounds rather than the fraction of a sharp identified set eliminated.

The index $\kappa(p, x)$ inherits its key properties directly from the nesting of expenditure bounds in Corollary 5.4 and the collapse characterization in Theorem 5.5.

Proposition 5.7 (Properties of the bound-tightening index). *Suppose the expenditure endpoints are finite and $W^M(p, x) > 0$. Then:*

- (i) $\kappa(p, x) \in [0, 1]$.
- (ii) $\kappa(p, x) > 0$ if and only if symmetry and homotheticity strictly tighten at least one expenditure bound:

$$\underline{e}^{SH}(p, x) > \underline{e}^M(p, x) \quad \text{or} \quad \bar{e}^{SH}(p, x) < \bar{e}^M(p, x).$$

By Corollary 5.4, this holds whenever a new structural comparison changes at least one of the two expenditure infima.

- (iii) $\kappa(p, x) = 1$ if and only if $\underline{e}^{SH}(p, x) = \bar{e}^{SH}(p, x)$: the frontier-contact condition of Theorem 5.5 holds and reference-contour expenditure is point identified within the maintained class.
- (iv) Holding the monotonicity baseline and the admissible utility class fixed, adding comparisons that are valid under the maintained structural assumptions can only weakly increase $\kappa(p, x)$. Adding observations generally changes both the structural and monotonicity intervals, so no unconditional monotonicity in sample size is asserted.

Proof. See Appendix A. □

Part (iv) formalizes the answer to the research question. The magnitude of identifying power delivered by symmetry and homotheticity is measured by $\kappa(p, x)$, which is observable from the data for any (p, x) : compute the monotone bounds from O , compute the symmetric-homothetic bounds from O_Σ , and take the ratio of widths. Because adding observations changes both the structural interval and the monotonicity benchmark, no general convergence of $\kappa(p, x)$ to one is implied. This gives empirical researchers a concrete diagnostic: for a given dataset and counterfactual price-bundle pair, by how much does the structural assumption contract the displayed valid interval?

6 Conclusion

Classical convexity-based revealed-preference welfare bounds can fail when the consumer’s true preferences are nonconvex. The monotonicity construction of Halevy et al. (2017) restores validity at the price of wider bounds. This paper shows that symmetry and homotheticity can tighten those reliable bounds through a distinct mechanism: symmetry adds all permutation-induced observations, while homotheticity adds valid radial comparisons.

A dataset is rationalizable by a symmetric and homothetic utility function if and only if its symmetrized expansion O_Σ satisfies HARP (Lemma 3.8). Violations can sometimes be detected through the pairwise PHARP screen, although passing that screen does not replace the full cyclic test. Conditional on the joint hypothesis, the corrected homothetic analogue of the monotone relation of Halevy et al. (2017) produces valid inner and outer bounds on every admissible strict upper contour set (Theorem 4.6). The corresponding reference-contour expenditure interval is uniformly valid over all continuous, strictly increasing, symmetric, and homothetic rationalizations, including nonconvex ones (Theorem 5.3).

The structural interval is weakly contained in the monotonicity-only interval and tightens strictly when a new permutation or radial comparison changes an expenditure-relevant frontier. If its two endpoints coincide, reference-contour expenditure is point identified within the maintained class (Theorem 5.5). The

bound-tightening index $\kappa(p, x)$ summarizes the proportional contraction of the displayed interval (Proposition 5.7). It is a diagnostic for this bounding procedure, not a claim about the fraction of a sharp identified set resolved.

The principal message is that nonconvexity need not leave welfare analysis at the monotonicity-only benchmark. Economically motivated and empirically testable restrictions can add valid revealed-preference comparisons without imposing convexity. Establishing sharpness would require a separate symmetric-homothetic order-extension theorem that constructs admissible rationalizations attaining all rankings not ruled out by the structural relation. Developing that theorem and taking the corrected bounds to household demand and index-number applications are natural directions for future research.

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A Proofs

Proof of Lemma 3.8

Normalize each observation by replacing p_t with $p_t/(p_t \cdot y_t)$, so $p_t \cdot y_t = 1$ without changing any budget. We prove $(i) \Rightarrow (ii) \Rightarrow (iii) \Leftrightarrow (iv)$ and $(iii) \Rightarrow (v) \Rightarrow (i)$.

$(i) \Rightarrow (ii)$. Immediate.

$(ii) \Rightarrow (iii)$. Suppose u is locally nonsatiated, homothetic, and symmetric and rationalizes O . If $p_t^\sigma \cdot x \leq p_t^\sigma \cdot y_t^\sigma$, then, writing $z = x^{\sigma^{-1}}$, permutation invariance of inner products gives $p_t \cdot z \leq p_t \cdot y_t$. Rationalization and symmetry imply $u(x) = u(z) \leq u(y_t) = u(y_t^\sigma)$. Thus u rationalizes every indexed observation in O_Σ . The necessity direction of the homothetic Afriat–Varian theorem then implies HARP on O_Σ .

$(iii) \Leftrightarrow (iv)$. This is the homothetic Afriat theorem applied to the indexed dataset O_Σ . In unnormalized units its inequalities are exactly those in part (iv); under the current normalization the denominators equal one.

$(iii) \Rightarrow (v)$. By the homogeneous Afriat–Varian construction, HARP on O_Σ implies the existence of a continuous, strictly increasing, concave, degree-one homogeneous utility v rationalizing O_Σ . Define

$$\bar{v}(x) := \frac{1}{|\Sigma|} \sum_{\rho \in \Sigma} v(x^\rho).$$

The function $\bar{v}$ retains all these properties and is symmetric. It rationalizes O_Σ : if $p_t^\sigma \cdot x \leq p_t^\sigma \cdot y_t^\sigma$, then for every ρ , $p_t^{\sigma\rho} \cdot x^\rho \leq p_t^{\sigma\rho} \cdot y_t^{\sigma\rho}$, so averaging the optimality inequalities for v gives $\bar{v}(x) \leq \bar{v}(y_t^\sigma)$.

Set $U_t := \bar{v}(y_t) > 0$. For fixed s, t, σ, η , put

$$c := \frac{p_t^\sigma \cdot y_s^\eta}{p_t^\sigma \cdot y_t^\sigma} > 0.$$

Then y_s^η/c lies on the budget of observation (t, σ) . Optimality gives $\bar{v}(y_s^\eta/c) \leq \bar{v}(y_t^\sigma) = U_t$. Homogeneity and symmetry yield $U_s = \bar{v}(y_s^\eta) \leq cU_t$, which is part (v).

(v) $\Rightarrow$ (i). Define

$$u(x) := \min_{t \in \mathbb{T}, \sigma \in \Sigma} \frac{U_t}{p_t \cdot y_t} p_t^\sigma \cdot x.$$

The function is continuous, concave, strictly increasing, and degree-one homogeneous. A permutation of x merely reindexes σ , so it is symmetric. Part (v) implies $u(y_t) = U_t$ for every t . If $p_t \cdot x \leq p_t \cdot y_t$, the term (t, id) gives $u(x) \leq U_t(p_t \cdot x)/(p_t \cdot y_t) \leq U_t = u(y_t)$. Thus u rationalizes O and part (i) holds. $\square$

Proof of Lemma 4.4

We first verify the utility implication of every direct link. If $x \succeq^M z$, monotonicity gives $u(x) \geq u(z)$, with strict inequality when the dominance is strict.

For a direct homothetic link, write $x = \lambda y_t$. The corrected affordability condition gives $p_t \cdot y_t \geq p_t \cdot (z/\lambda)$, so rationalization implies $u(y_t) \geq u(z/\lambda)$. Homotheticity then gives $u(x) = u(\lambda y_t) \geq u(z)$. When the affordability inequality is strict, z/λ lies strictly inside the observed budget. Local nonsatiation and rationalization imply $u(y_t) > u(z/\lambda)$; homotheticity preserves this strict ranking and therefore $u(x) > u(z)$.

Now follow any chain in Definition 2.6. Every weak link yields a weak utility inequality and every strict link yields a strict inequality. Because a strict chain contains at least one strict link, concatenating the inequalities gives $x \succ_H^{M,R} z \Rightarrow u(x) > u(z)$, regardless of where the strict link occurs. $\square$

Proof of Proposition 3.11

Proof. Let

$$O_\Sigma = \{(p_t^\sigma, y_t^\sigma) : (t, \sigma) \in \mathbb{T} \times \Sigma\}$$

denote the symmetrized dataset. By Lemma 3.8, the original dataset O is rationalizable by a continuous, strictly increasing, concave, homothetic, and symmetric utility function if and only if O_Σ satisfies HARP.

We prove each part in turn.

(i) Sufficiency for refutation. PHARP on O_Σ requires that, for every $s, t \in \mathbb{T}$ and every $\sigma, \eta \in \Sigma$,

$$(p_t^\sigma y_s^\eta)(p_s^\eta y_t^\sigma) \geq (p_t^\sigma y_t^\sigma)(p_s^\eta y_s^\eta).$$

Suppose PHARP is violated. Then there exist $s, t \in \mathbb{T}$ and $\sigma, \eta \in \Sigma$ such that

$$(p_t^\sigma y_s^\eta)(p_s^\eta y_t^\sigma) < (p_t^\sigma y_t^\sigma)(p_s^\eta y_s^\eta).$$

This is exactly a length-two violation of HARP on the symmetrized dataset O_Σ . Hence O_Σ does not satisfy HARP. By Lemma 3.8, no continuous, strictly increasing, concave, homothetic, and symmetric utility function can rationalize O .

(ii) Permutation reduction. Adopt the convention $y^\sigma = (y_{\sigma(1)}, \dots, y_{\sigma(L)})$. For any $\sigma, \eta \in \Sigma$,

$$p_t^\sigma \cdot y_s^\eta = \sum_{i=1}^L p_{t,\sigma(i)} y_{s,\eta(i)} = \sum_{j=1}^L p_{t,j} y_{s,\eta(\sigma^{-1}(j))} = p_t \cdot y_s^{\eta \circ \sigma^{-1}},$$

and by the same computation with the roles of (t, σ) and (s, η) exchanged, $p_s^\eta \cdot y_t^\sigma = p_s \cdot y_t^{\sigma \circ \eta^{-1}}$. Setting $\pi := \eta \circ \sigma^{-1}$ gives $\sigma \circ \eta^{-1} = \pi^{-1}$. Finally $p_t^\sigma \cdot y_t^\sigma = \sum_i p_{t,\sigma(i)} y_{t,\sigma(i)} = p_t \cdot y_t$, and likewise for s . Substituting into the PHARP inequality yields (1). The left-hand side depends on (σ, η) only through π , and as (σ, η) ranges over $\Sigma \times \Sigma$ the composition π ranges over all of Σ ; the $(L!)^2$ inequalities therefore reduce to $L!$ distinct ones, and $T^2 L!$ in total across ordered pairs.

(iii) Rearrangement screen. Both factors in (1) are strictly positive, so for every π ,

$$(p_t \cdot y_s^\pi)(p_s \cdot y_t^{\pi^{-1}}) \geq m(p_t, y_s) m(p_s, y_t),$$

whence the left-hand minimum in (1) is bounded below by $m(p_t, y_s)m(p_s, y_t)$, giving (a). Statement (b) is immediate, since the minimum over Σ is no larger than the value at any particular π . That $m(p, y)$ is attained by pairing p in ascending order with y in descending order is the rearrangement inequality; sorting both vectors costs

$O(L \log L)$.

(iv) Computational implication. Let $N := |O_\Sigma| = TL!$. By part (ii), PHARP requires checking $T^2L! = N^2/(T(L!)) \cdot L!$ inequalities in the worst case, which is polynomial in N though factorial in the primitive number of goods. By part (iii) each ordered pair is resolved in $O(L \log L)$ whenever either screen applies, so the factorial cost is incurred only on pairs falling between the two screens. Because a single violation is sufficient to reject the joint symmetry–homotheticity hypothesis, PHARP provides a one-sided refutation procedure.

Finally, this procedure is no more demanding in structure than standard pairwise revealed-preference diagnostics: it requires only pairwise comparisons between observations, rather than the solution of a global system of revealed-preference inequalities. This proves the result. $\square$ $\square$

Proof of Lemma 4.5

If $y \in MRP_{O_\Sigma}^H(x)$, then $y \succ_H^{M,R} x$, so by Lemma 4.4 we have $u(y) > u(x)$, i.e., $y \in P_u(x)$. Hence $MRP_{O_\Sigma}^H(x) \subseteq P_u(x)$.

If $y \notin NMRW_{O_\Sigma}^H(x)$, then $y \in MRW_{O_\Sigma}^H(x)$, i.e., $x \succ_H^{M,R} y$. By Lemma 4.4, $u(x) > u(y)$, so $y \notin P_u(x)$. Hence $P_u(x) \subseteq NMRW_{O_\Sigma}^H(x)$. Combining: $MRP_{O_\Sigma}^H(x) \subseteq P_u(x) \subseteq NMRW_{O_\Sigma}^H(x)$. $\square$

Proof of Proposition 4.2

Step 1: $MRP_O(x) \subseteq MRP_{O_\Sigma}(x)$. Since $O \subseteq O_\Sigma$, every monotone revealed-preference chain in O is also in O_Σ .

Step 2: $NMRW_{O_\Sigma}(x) \subseteq NMRW_O(x)$. Symmetrisation adds revealed-preference chains but never removes them. Hence $MRW_O(x) \subseteq MRW_{O_\Sigma}(x)$, and taking complements yields the result.

Step 3: $MRP_{O_\Sigma}(x) \subseteq P_u(x) \subseteq NMRW_{O_\Sigma}(x)$. If $y \succ_H^{M,R} x$ in O_Σ , then since u rationalizes O_Σ and u is monotone, a standard Afriat argument gives $u(y) > u(x)$, so $y \in P_u(x)$. Conversely, if $y \notin NMRW_{O_\Sigma}(x)$ then $x \succ_H^{M,R} y$ in O_Σ , so $u(x) > u(y)$ and $y \notin P_u(x)$.

Strictness. If symmetrization generates $y \succ^{M,R} x$ in O_Σ but not in O , then $y \in MRP_{O_\Sigma}(x) \setminus MRP_O(x)$, making the first inclusion strict. $\square$

Proof of Proposition 4.3

Ordinary direct revealed preference is the special case $\lambda = 1$ of the corrected direct homothetic relation. The homothetic construction uses the same monotone first-link option as the Halevy baseline. Therefore every monotonic revealed-preference chain on O is also a valid monotone-homothetic chain. Hence $MRP_O(x) \subseteq MRP_O^H(x)$ and $MRW_O(x) \subseteq MRW_O^H(x)$. Taking complements gives $NMRW_O^H(x) \subseteq NMRW_O(x)$. A radial chain that is absent from the baseline makes the corresponding inclusion strict. $\square$

Proof of Theorem 4.6

Step 1: $MRP_O^H(x) \subseteq MRP_{O_\Sigma}^H(x)$. Since $O \subseteq O_\Sigma$, every monotone-homothetic chain in O is in O_Σ .

Step 2: $NMRW_{O_\Sigma}^H(x) \subseteq NMRW_O^H(x)$. Since $O \subseteq O_\Sigma$, the larger dataset reveals more bundles to be worse than x : $MRW_O^H(x) \subseteq MRW_{O_\Sigma}^H(x)$. Taking complements gives the result.

Step 3: $MRP_{O_\Sigma}^H(x) \subseteq P_u(x) \subseteq NMRW_{O_\Sigma}^H(x)$. Since u is monotone, homothetic, and symmetric and rationalizes O_Σ , Lemma 4.4 applied to O_Σ gives: if $y \succ_H^{M,R} x$ in O_Σ then $u(y) > u(x)$, so $y \in P_u(x)$. If $y \notin NMRW_{O_\Sigma}^H(x)$ then $x \succ_H^{M,R} y$ in O_Σ , so $u(x) > u(y)$ and $y \notin P_u(x)$. $\square$

Proof of Proposition 4.8

Since $O \subseteq O_\Sigma$, every monotonic chain available in O is available in O_Σ . By Proposition 4.3, every monotonic chain on O_Σ is also a valid monotone-homothetic chain. Hence

$$MRP_O(x) \subseteq MRP_{O_\Sigma}^H(x), \quad MRW_O(x) \subseteq MRW_{O_\Sigma}^H(x).$$

Taking complements gives $NMRW_{O_\Sigma}^H(x) \subseteq NMRW_O(x)$.

If $y \in ID_{O_\Sigma}^H(x)$, then $y \in NMRW_{O_\Sigma}^H(x) \subseteq NMRW_O(x)$ and $y \notin MRP_{O_\Sigma}^H(x)$. Since $MRP_O(x) \subseteq MRP_{O_\Sigma}^H(x)$, this implies $y \notin MRP_O(x)$. Therefore $y \in ID_O(x)$, proving $ID_{O_\Sigma}^H(x) \subseteq ID_O(x)$. $\square$

Proof of Proposition 5.2

By Theorem 4.6, $MRP_{O_\Sigma}^H(x) \subseteq P_u(x) \subseteq NMRW_{O_\Sigma}^H(x)$. For nested sets, minimizing a fixed objective over the larger set gives the smaller infimum. Thus

$$\underline{e}(p, x) \leq \inf_{z \in P_u(x)} p \cdot z \leq \bar{e}(p, x).$$

Continuity and strict monotonicity imply $\inf_{z \in P_u(x)} p \cdot z = e_u(p; x)$: any weakly preferred bundle can be perturbed upward by an arbitrarily small positive vector to obtain a strictly preferred bundle at arbitrarily close expenditure. This proves the result. $\square$

Proof of Theorem 5.3

Proposition 5.2 gives $\underline{e}(p, x) \leq e_u(p; x) \leq \bar{e}(p, x)$ for every $u \in \mathcal{U}$. Taking the infimum and supremum over $\mathcal{U}$ yields the displayed chain. No attainability claim is used or implied. $\square$

Proof of Corollary 5.4

Proposition 4.3 and Theorem 4.6 give

$$MRP_O(x) \subseteq MRP_O^H(x) \subseteq MRP_{O_\Sigma}^H(x)$$

and

$$NMRW_{O_\Sigma}^H(x) \subseteq NMRW_O^H(x) \subseteq NMRW_O(x).$$

For $A \subseteq B$, $\inf_B p \cdot z \leq \inf_A p \cdot z$. Applying this rule to the two chains and combining with Proposition 5.2 gives the displayed expenditure hierarchy. Strictness requires a strict set change that moves the relevant infimum; a new chain that is not cost-active

need not tighten an expenditure endpoint. $\square$

Proof of Theorem 5.5

Part (i) follows from Proposition 5.2: every admissible $e_u(p; x)$ lies between the two endpoints, so equality of the endpoints pins down their common value.

For part (ii), let z^* minimize $p \cdot z$ over $NMRW_{O_\Sigma}^H(x)$ and suppose $z^* \in \overline{MRP_{O_\Sigma}^H(x)}$. A sequence from the inner set converges to z^* , so continuity of the linear objective gives $\bar{e}_\Sigma(p, x) \leq p \cdot z^* = \underline{e}_\Sigma(p, x)$. The reverse inequality always holds because $MRP_{O_\Sigma}^H(x) \subseteq NMRW_{O_\Sigma}^H(x)$.

For part (iii), the lower-support condition gives $\underline{e}_\Sigma(p, x) \geq e^*$, while the approximating inner bundles give $\bar{e}_\Sigma(p, x) \leq e^*$. Again the general ordering $\underline{e}_\Sigma \leq \bar{e}_\Sigma$ forces equality. $\square$

Proof of Proposition 5.7

Part (i). By Corollary 5.4, the structural interval is contained in the monotonicity interval. Hence $0 \leq W^{SH}(p, x) \leq W^M(p, x)$ and $\kappa(p, x) \in [0, 1]$.

Part (ii). The index is positive exactly when the nested structural interval has strictly smaller width, which is equivalent to at least one endpoint moving strictly inward.

Part (iii). The index equals one exactly when $W^{SH}(p, x) = 0$, i.e. when the two structural endpoints coincide. Theorem 5.5 then gives point identification within the maintained class.

Part (iv). Hold the monotonicity baseline and admissible class fixed. Adding valid structural comparisons can only enlarge the structural revealed-preferred and revealed-worse sets and therefore shrink the not-revealed-worse set. The structural lower endpoint can only rise and the upper endpoint can only fall, so W^{SH} weakly decreases while W^M is unchanged. If new observations also alter the monotonicity baseline, both numerator and denominator change and no general monotonicity follows. $\square$

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